EXPERIMENT 10 : MUSIC APP DATA VISUALIZATION

AIM:

To create data visualizations such as pie charts and bar graphs to analyze music listening patterns using JavaScript and Chart.js.

OBJECTIVE:

- To understand data visualization techniques
- To represent user listening data using charts
- To use Chart.js for creating interactive graphs
- To display values inside a pie chart

INTRODUCTION:

Data Visualization is the graphical representation of data using charts and graphs. In this experiment, a **music analytics system (BeatFlow)** is developed to visualize listening data such as time spent on different genres like Pop, Rock, Hip-Hop, Classical, and EDM.

PROCEDURE:

Step 1: Create HTML File (index.html)

- Create a basic HTML structure
- Add canvas elements for charts
- Include Chart.js and datalabels plugin

Step 2: Create JavaScript File (script.js)

- Define music listening data
- Configure pie chart with values inside
- Configure bar chart for comparison

Step 3: Run the Project

- Open index.html in a browser

CODE:

index.html

```
<!DOCTYPE html>

<html lang="en">

<head>

   <meta charset="UTF-8">

   <meta name="viewport" content="width=device-width, initial-scale=1.0">

   <title>BeatFlow Music Visualization</title>

   <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>

   <script
src="https://cdn.jsdelivr.net/npm/chartjs-plugin-datalabels"></script>

   <style>

       * {

           margin: 0;

           padding: 0;

           box-sizing: border-box;

       }

       body {

           font-family: -apple-system, BlinkMacSystemFont, 'Segoe UI', Roboto,
sans-serif;

           background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);

           min-height: 100vh;

           padding: 20px;
```

```
        color: #333;
    }
    .container {
        max-width: 1200px;
        margin: 0 auto;
    }
    h1 {
        text-align: center;
        color: white;
        margin-bottom: 40px;
        font-size: 2.5rem;
        text-shadow: 0 2px 10px rgba(0,0,0,0.3);
    }
    .charts-grid {
        display: grid;
        grid-template-columns: repeat(auto-fit, minmax(400px, 1fr));
        gap: 30px;
        margin-bottom: 20px;
    }
    .chart-card {
        background: rgba(255,255,255,0.95);
        border-radius: 20px;
        padding: 30px;
        box-shadow: 0 20px 40px rgba(0,0,0,0.1);
        backdrop-filter: blur(10px);
        transition: transform 0.3s ease;
```

```
}
.chart-card:hover {
    transform: translateY(-5px);
}
.chart-card h2 {
    text-align: center;
    margin-bottom: 20px;
    color: #4a5568;
    font-size: 1.5rem;
}
canvas {
    max-width: 100%;
    max-height: 400px;
    width: 400px !important;
    height: 400px !important;
    display: block;
    margin: 0 auto;
}
@media (max-width: 768px) {
    .charts-grid {
        grid-template-columns: 1fr;
        gap: 20px;
    }
    canvas {
        width: 300px !important;
        height: 300px !important;
```

```
            }
            h1 {
                font-size: 2rem;
            }
        }
    </style>
</head>
<body>
    <div class="container">
        <h1>🎵 BeatFlow Music Visualization</h1>
        <div class="charts-grid">
            <div class="chart-card">
                <h2>Genre Distribution</h2>
                <canvas id="pieChart"></canvas>
            </div>
            <div class="chart-card">
                <h2>Listening Frequency</h2>
                <canvas id="barChart"></canvas>
            </div>
        </div>
    </div>
    <script src="script.js"></script>
</body>
</html>
```

script.js

```
// Pie Chart (Genre Distribution)

const pieData = {

   labels: ['Pop', 'Rock', 'Hip-Hop', 'Classical', 'EDM'],

   datasets: [{

       data: [30, 20, 25, 10, 15],

       backgroundColor: [

           '#FF6384',

           '#36A2EB',

           '#FFCE56',

           '#4BC0C0',

           '#9966FF'

       ]

   }]

};



new Chart(document.getElementById('pieChart'), {

   type: 'pie',

   data: pieData,

   plugins: [ChartDataLabels],

   options: {

       responsive: true,

       maintainAspectRatio: false,

       plugins: {

           datalabels: {

               color: '#fff',
```

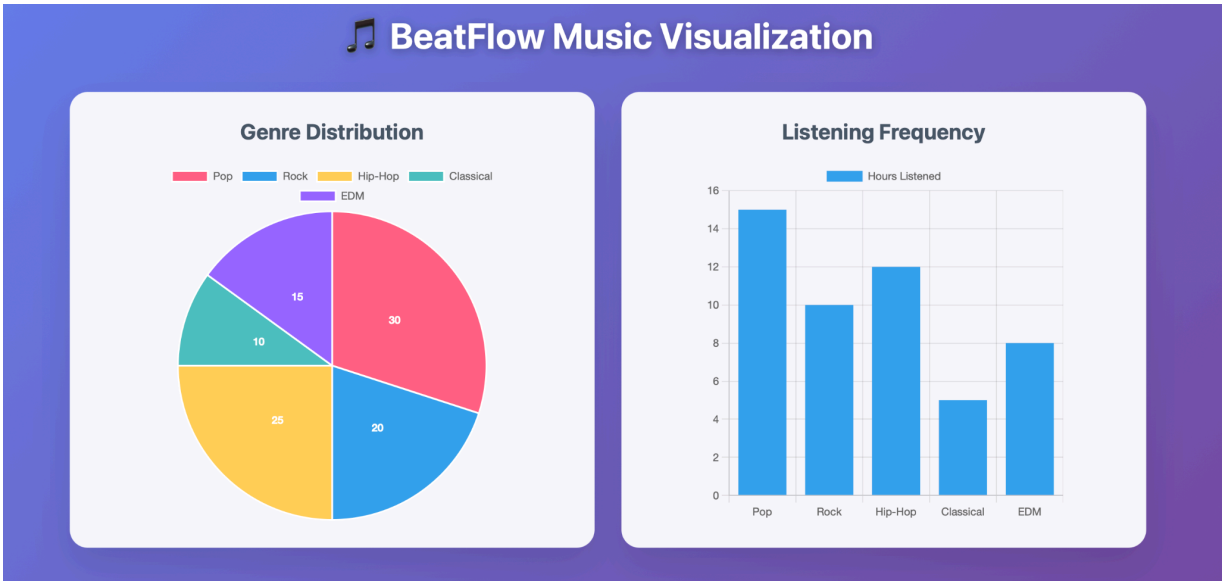
```
                font: {
                    weight: 'bold'
                }
            }
        }
    }
});



// Bar Chart (Listening Frequency)
const barData = {
    labels: ['Pop', 'Rock', 'Hip-Hop', 'Classical', 'EDM'],
    datasets: [{
        label: 'Hours Listened',
        data: [15, 10, 12, 5, 8],
        backgroundColor: '#36A2EB'
    }]
};


new Chart(document.getElementById('barChart'), {
    type: 'bar',
    data: barData,
    options: {
        responsive: true,
        maintainAspectRatio: false
    }
```

```
});
```

Refer to index.html and script.js files created using Chart.js and datalabels plugin.

OUTPUT:

- Pie chart showing distribution of time spent on music genres
- Bar chart comparing listening frequency of different genres



RESULT:

The music data visualization was successfully created using JavaScript and Chart.js, effectively representing user listening patterns through pie and bar charts.